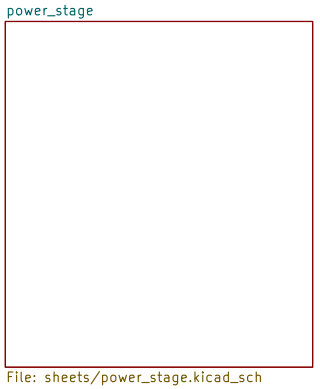
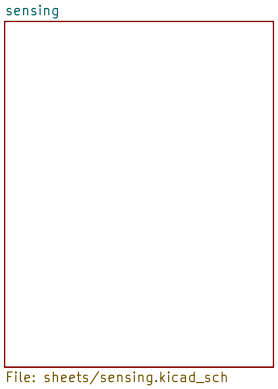
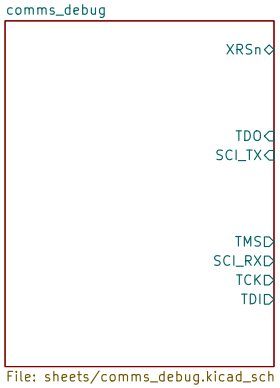
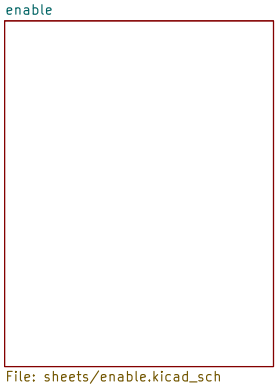






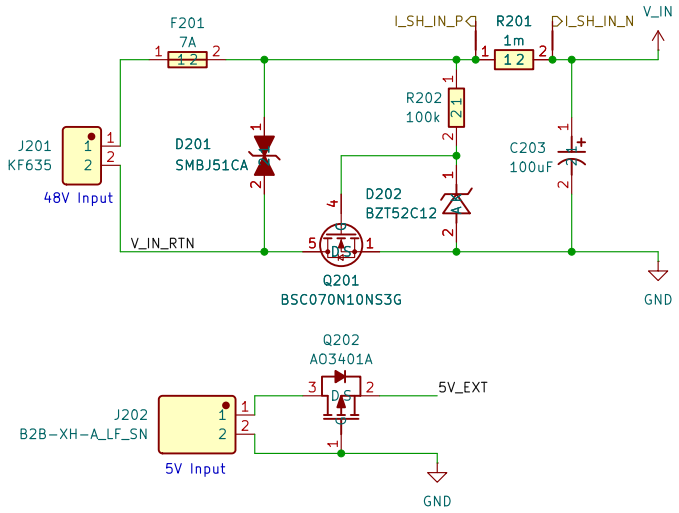




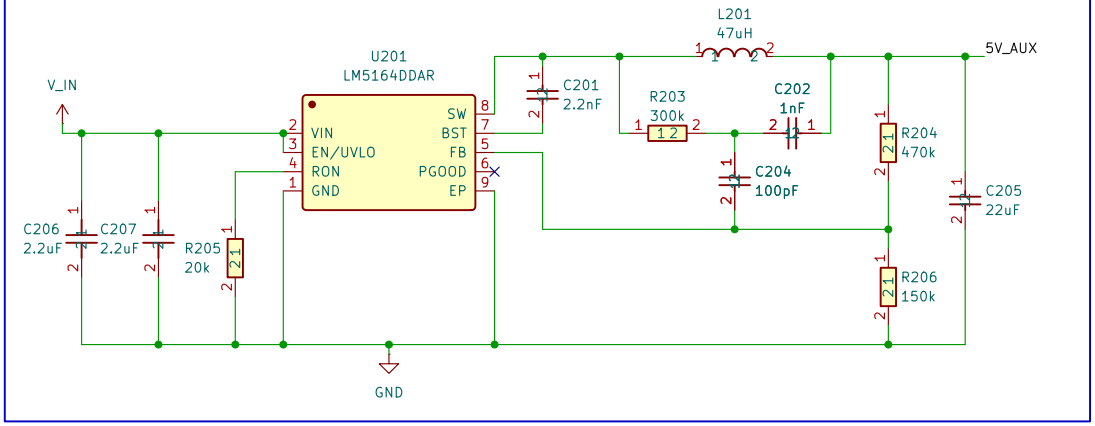


Owner: [NAME]
HexaGaN
Sheet: /
File: gan-buck-converter.kicad_sch
Title: 48 V to 12 V GaN Buck – Top
Size: A4 | Date: | **Rev: 0.1**
KiCad E.D.A. 10.0.6 | Id: 1/7

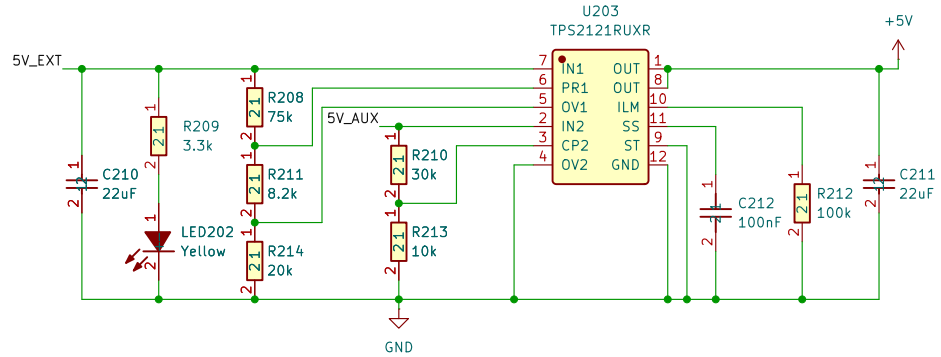
Input Entry and Protections



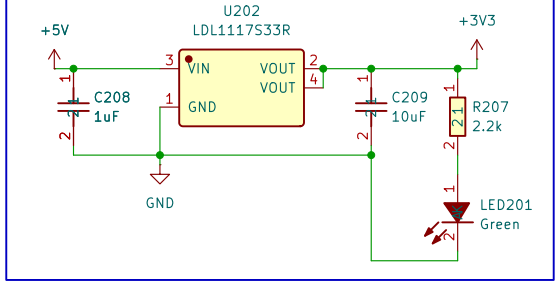
Aux Regulator (V_IN -> 5V_AUX)



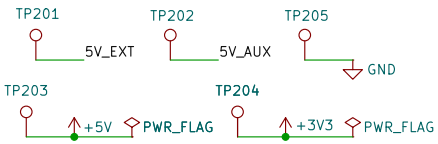
Power MUX



Linear Regulator (5V -> 3V3)



Test Points

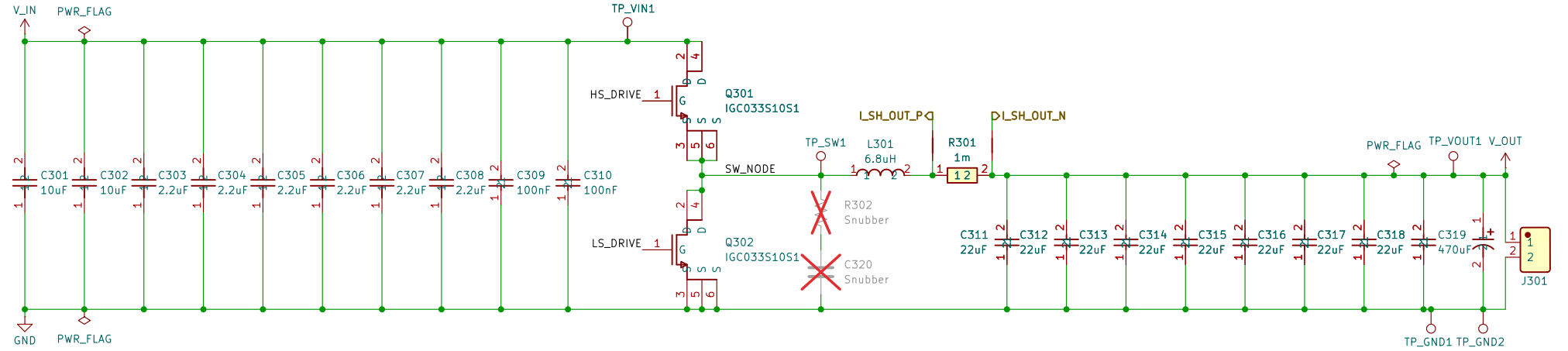


Owner: Reza Rahimzadeh
HexaGaN
Sheet: /input_supply/
File: input_supply.kicad_sch

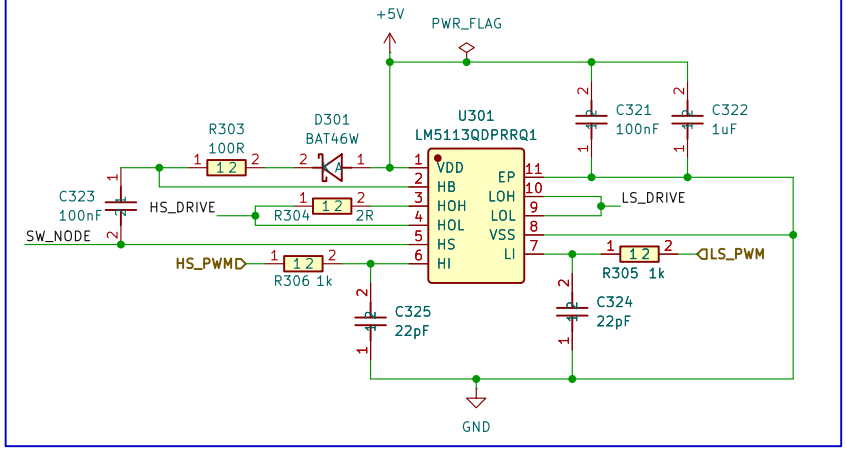
Title: Input Supply

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/7

Synchronous Buck Converter



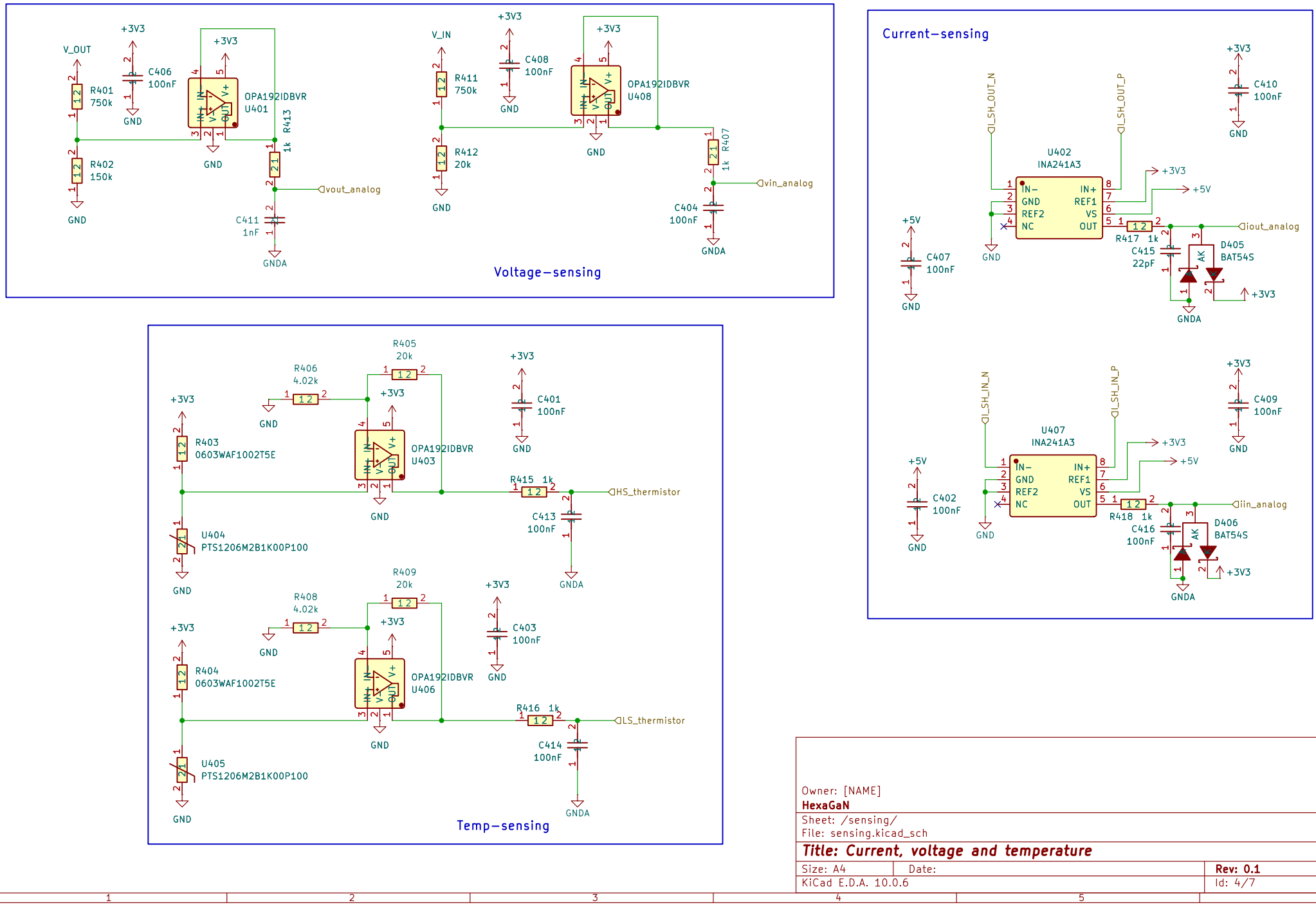
Gate Driver

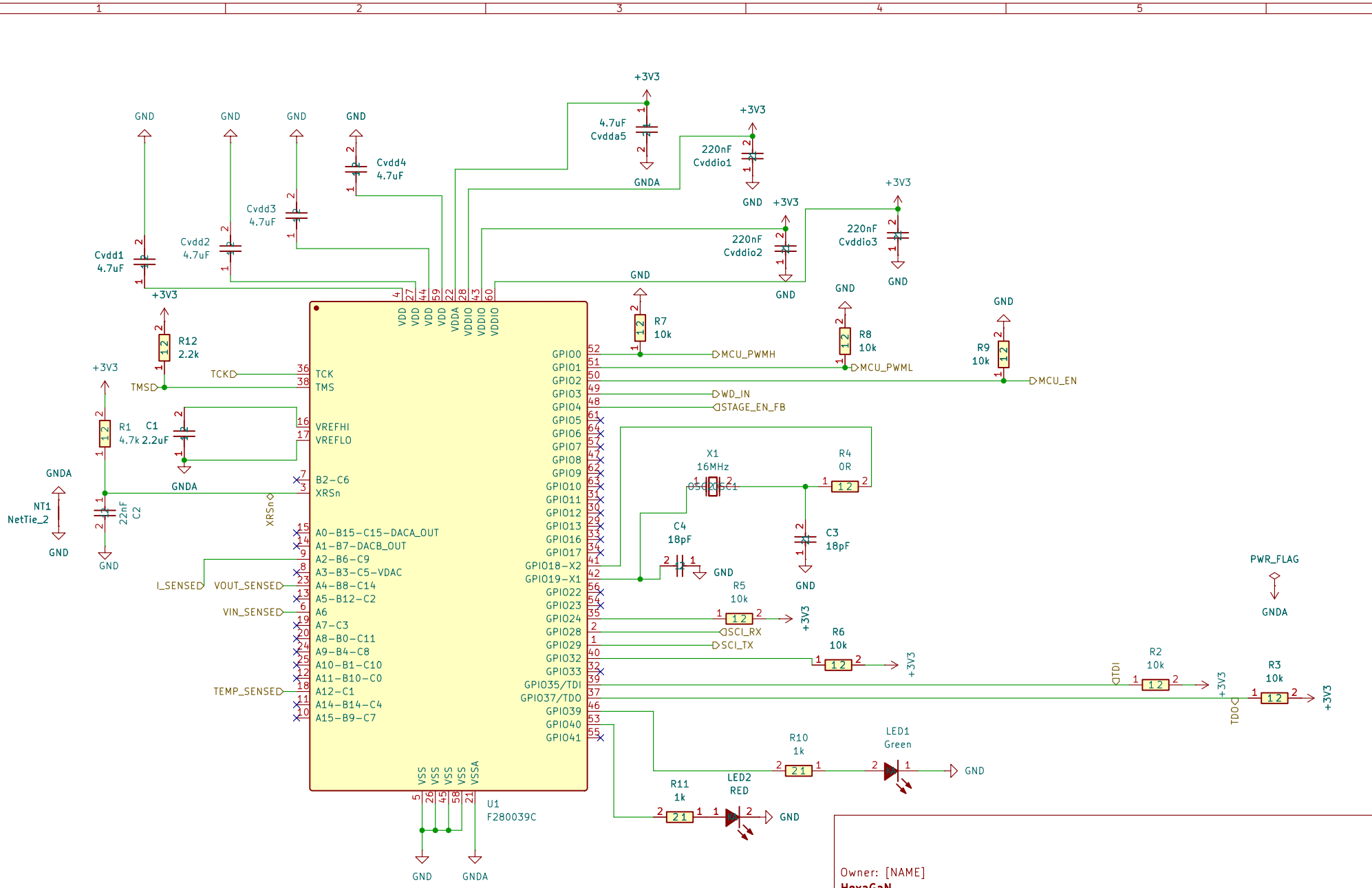


Owner: Reza Rahimzadeh
HexaGaN
Sheet: /power_stage/
File: power_stage.kicad_sch

Title: Power Stage

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 3/7

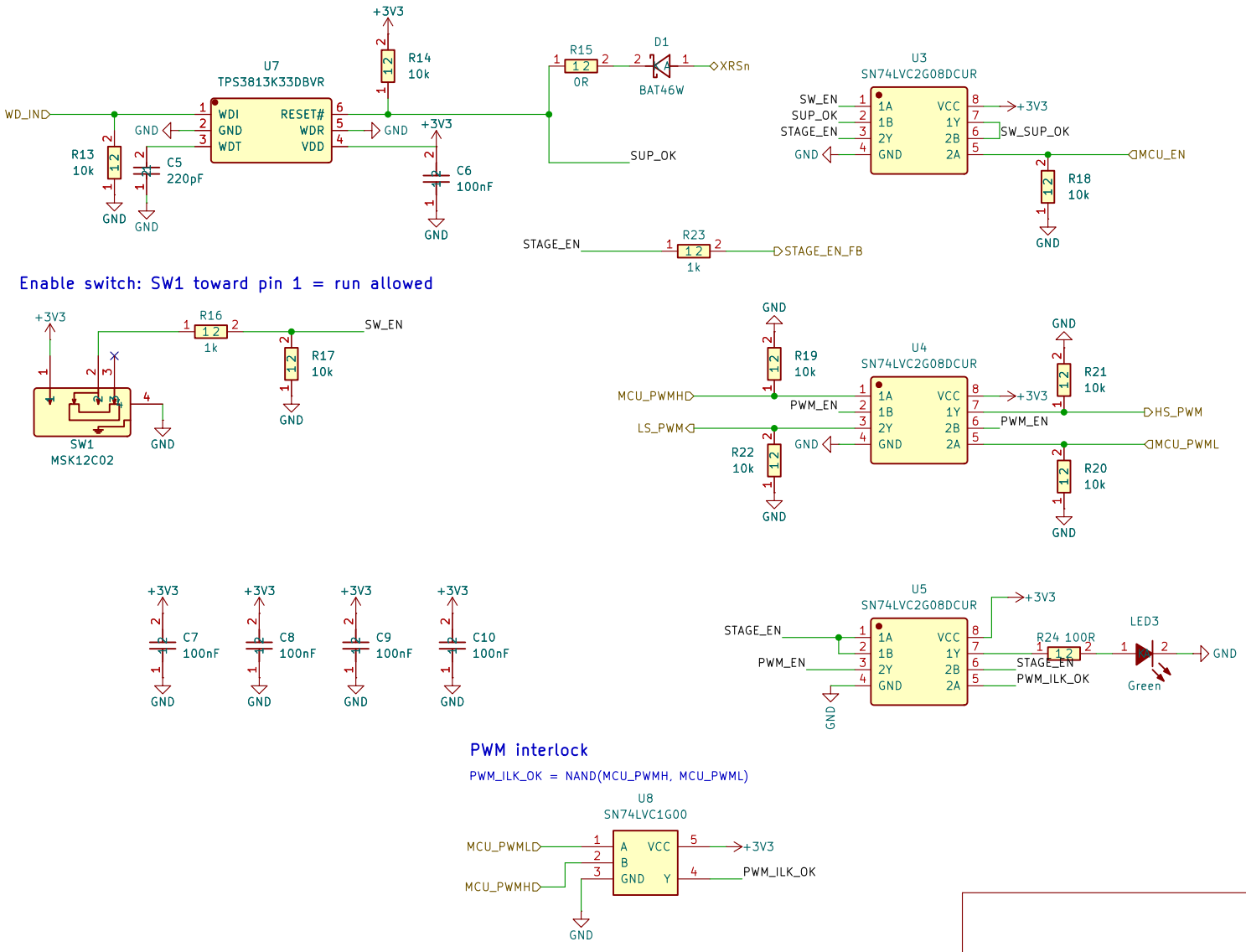

Voltage−sensing
V_OUT
R401 750k
R402 150k
C406 100nF
OPA192IDBVR U401
R413 1k
C411 1nF
vout_analog
V_IN
R411 750k
R412 20k
C408 100nF
OPA192IDBVR U408
R407 1k
C404 100nF
vin_analog
Current−sensing
U402 INA241A3
C410 100nF
C407 100nF
R417 1k
C415 22pF
D405 BAT54S
iout_analog
U407 INA241A3
C409 100nF
C402 100nF
R418 1k
C416 100nF
D406 BAT54S
iin_analog
Temp−sensing
R405 20k
R406 4.02k
R403 0603WAF1002T5E
U404 PTS1206M2B1K00P100
OPA192IDBVR U403
C401 100nF
R415 1k
C413 100nF
HS_thermistor
R409 20k
R408 4.02k
R404 0603WAF1002T5E
U405 PTS1206M2B1K00P100
OPA192IDBVR U406
C403 100nF
R416 1k
C414 100nF
LS_thermistor
Owner: [NAME]
HexaGaN
Sheet: /sensing/
File: sensing.kicad_sch
Title: Current, voltage and temperature
Size: A4
Date:
Rev: 0.1
KiCad E.D.A. 10.0.6
Id: 4/7



Owner: [NAME]
HexaGaN
Sheet: /mcu/
File: mcu.kicad_sch

Title: Microcontroller (U1)

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/7

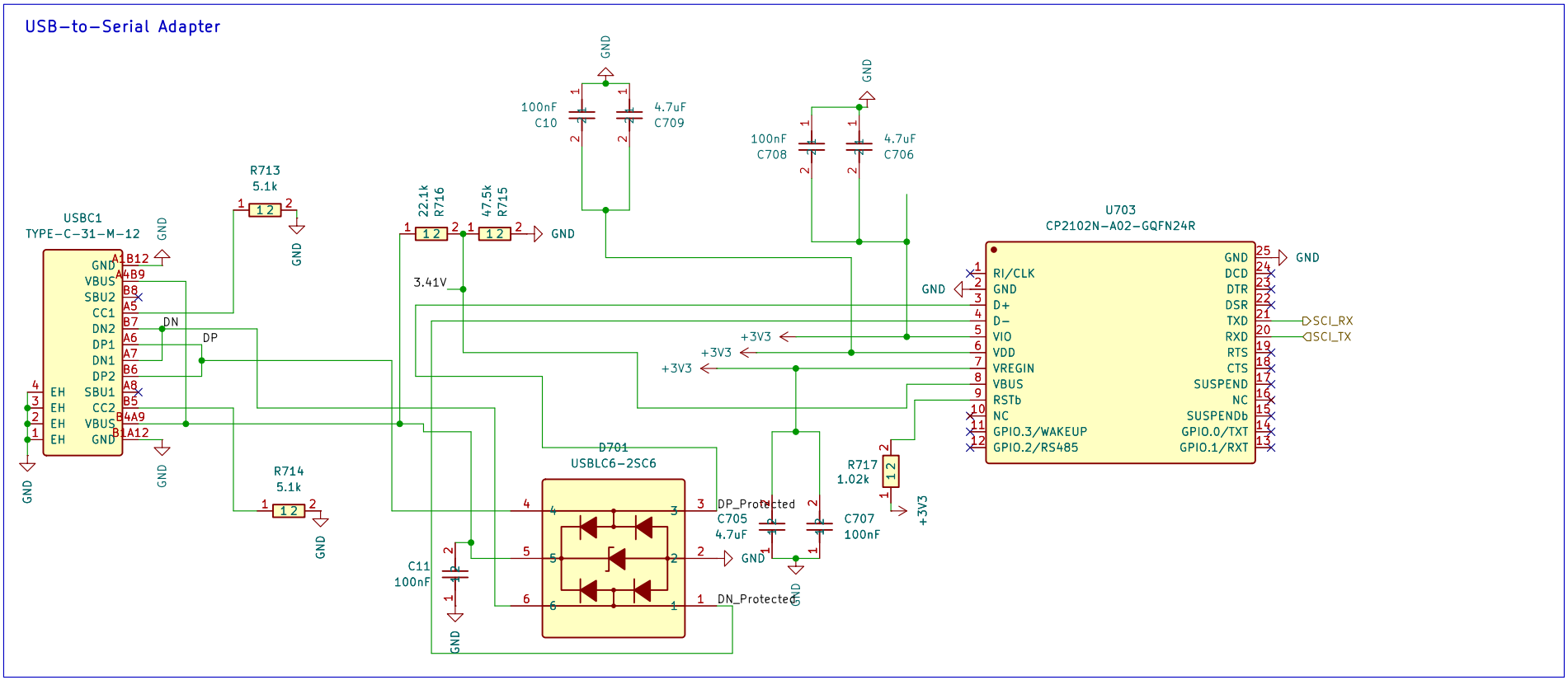


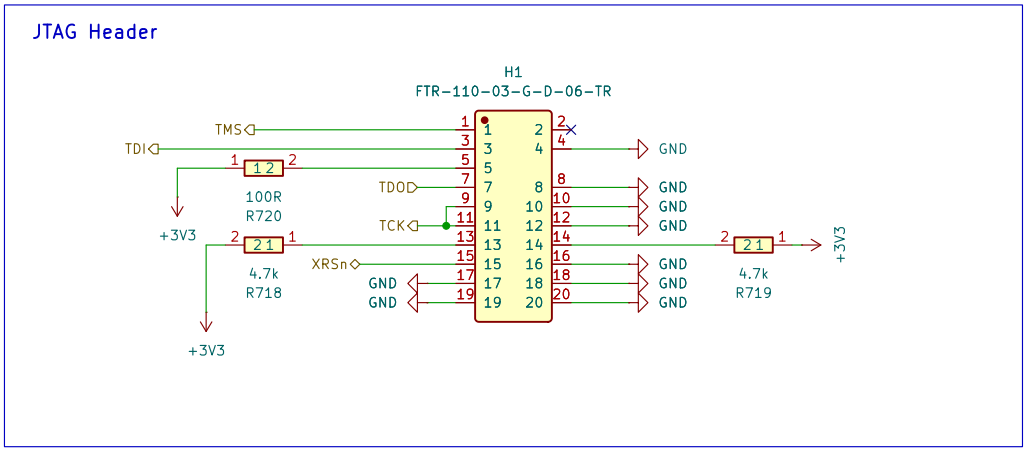
Owner: [NAME]
HexaGaN

Sheet: /enable/
File: enable.kicad_sch

Title: Hardware enable and watchdog

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 6/7





Owner: [NAME]	
HexaGaN	
Sheet: /comms_debug/	
File: comms_debug.kicad_sch	
Title: USB serial and JTAG	
Size: A4	Date:
KiCad E.D.A. 10.0.6	Id: 7/7
	Rev: 0.1